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Step-by-step solution

Step 1 of 3

The given information entails that the income earned by Jerry is \$12. With a price of \$2 per unit of yogurt, the maximum number of units of yogurt he can purchase is 6. Similarly, he can consume 3 berries at a price of \$4 per unit with his given income.

Comment

Step 2 of 3

A bundle is affordable when it can satisfy the budget constraint:

$$M \geq xPx + yPy$$

where

$$M = \text{income}$$
$$x = \text{good } x$$
$$y = \text{good } y$$
$$Px = \text{price of } x$$
$$Py = \text{price of } y$$

Substitute the values of the consumption in the desirable bundle, price of two goods and the given income to check whether the desirable bundle is affordable.

$$M \geq xPx + yPy$$
$$12 \geq x \times 2 + y \times 4$$
$$12 \geq 4x + 2y$$

Hence the budget constraint suggests that all bundles of berries and yogurt that satisfy  $12 \geq 4x + 2y$ , will be affordable.

Comment

Step 3 of 3

The graph of Jerry's budget line is shown in Figure 1. In this case, the current income is sufficient to buy the selected four bundles so **that it is affordable and it lies on the budget line.**

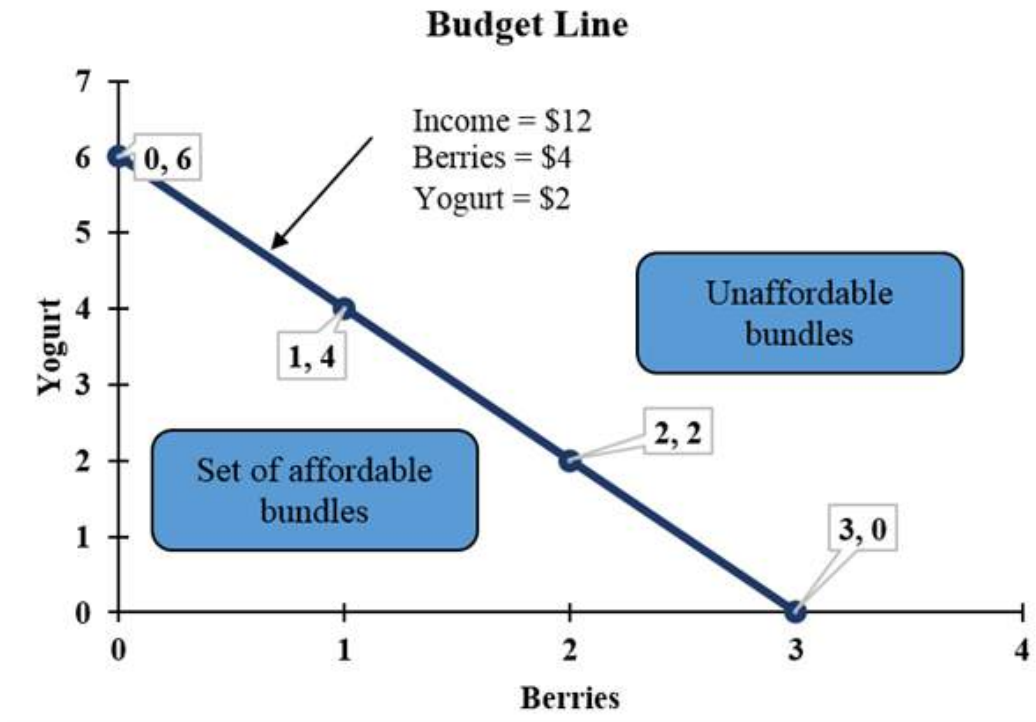


Figure 1

With  $12 \geq 4x + 2y$ , the affordable bundles for berries and yogurt are: 0 units of berries and 6 units of yogurt, 1 units of berries and 4 units of yogurt, 2 units of berries and 2 units of yogurt and 3 units of berries and 0 units of yogurt.

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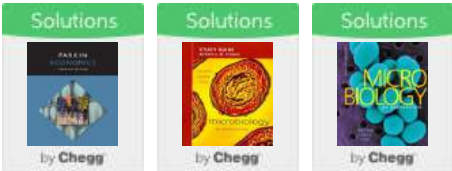
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Step-by-step solution

Step 1 of 2

a)

When the price of berries falls from \$4 to \$2, the original budget constraint rotates around the vertical intercept (since the price of yogurt remains unchanged). This is illustrated in Figure 1.

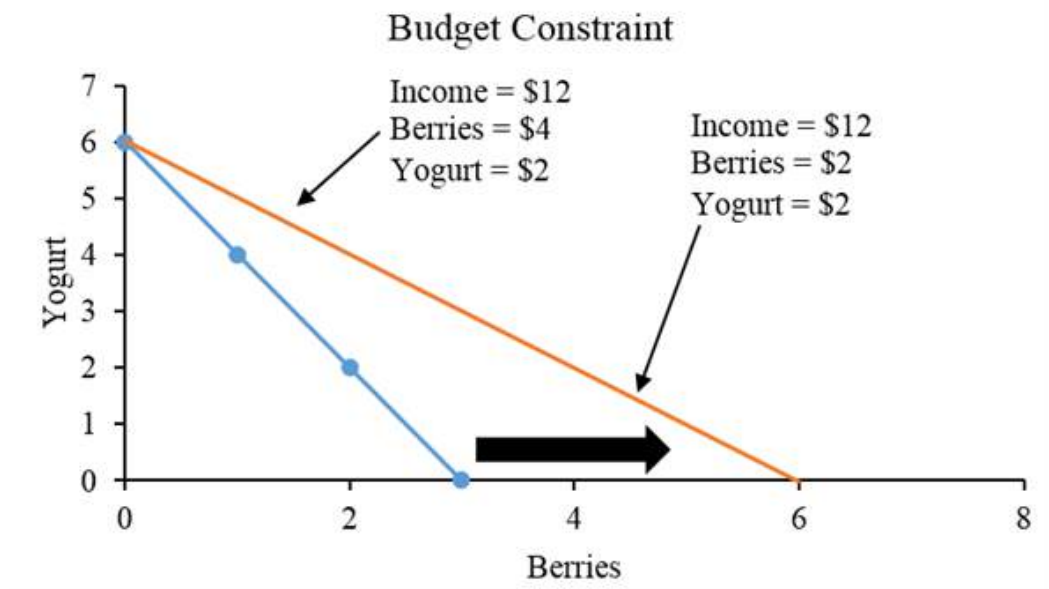


Figure 1

The new horizontal intercept allows him to purchase a maximum of 6 berries. The new budget line (orange one) lies outside the original one, as shown by Figure 1.

Comment

Step 2 of 2

b)

With the rise in income, Jerry can now consume more of both the goods. This implies his budget line shifts out to the right. Both vertical and horizontal intercepts are increased, exhibiting a higher real purchasing power.

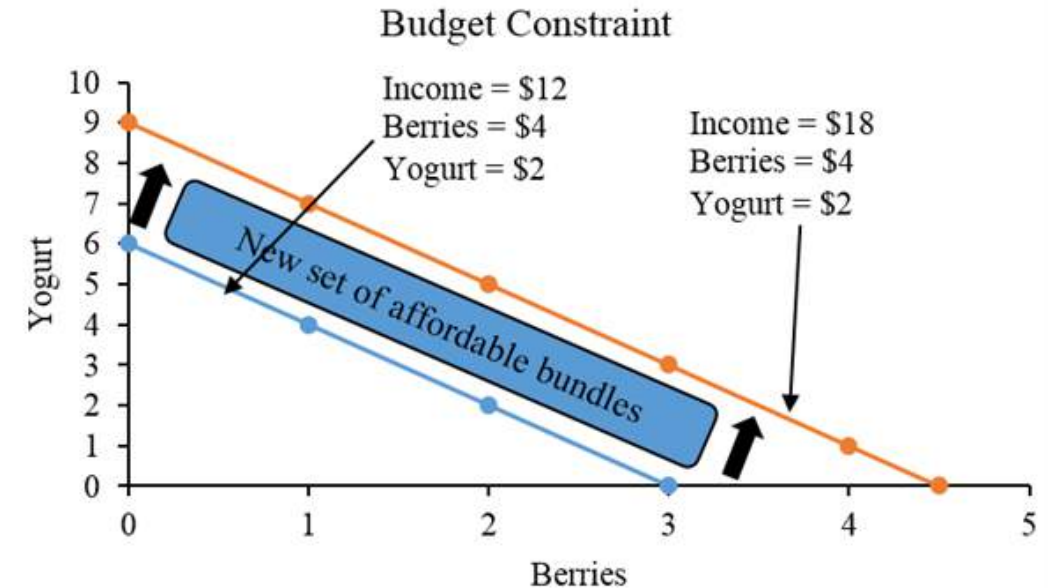


Figure 2

The new budget line has income increased to \$18. Jerry can now consume more of both the goods, again presuming them as normal goods. The substitution effect might not dominate the income effect and in the end he consumes more of both. This income gives him an entire new yet affordable bundles.

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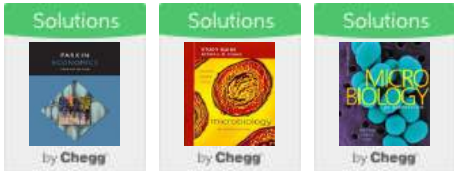
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## Step-by-step solution

### Step 1 of 4

Marginal utility is defined as change in total utility due to one unit increase in consumption of that commodity. Mathematically marginal utility is determined by the following formula.

Suppose, consumption increases from  $X_1$  to  $X_2$  and due to this increment total utility changes from  $U(X_1)$  to  $U(X_2)$ , then marginal utility ( $MU_x$ ) is defined as:

$$\begin{aligned} MU_x &= \frac{U(X_2) - U(X_1)}{X_2 - X_1} \\ &= \frac{\Delta U(X)}{\Delta X} \end{aligned}$$

Comment

### Step 2 of 4

As marginal utility is a measure that incorporates change in total utility we can not get it for first unit of consumption. From second unit we can calculate change in total utility due to one unit more consumption. In this problem second unit's total utility ( $U(X_2)$ ) is 220 unit and first unit's utility is ( $U(X_1)$ ) 120 unit.

Then change in total utility  $= U(X_2) - U(X_1)$

$$= 220 - 120$$

$$= 100$$

Change in consumption  $= (X_2 - X_1)$

$$= (2 - 1)$$

$$= 1$$

Comment

### Step 3 of 4

Now Marginal utility of second hour is

$$\begin{aligned} MU_2 &= \frac{U(X_2) - U(X_1)}{X_2 - X_1} \\ &= \frac{100}{1} \\ &= 100 \end{aligned}$$

In this same manner we get marginal utility up to 7<sup>th</sup> hour. Following table calculates the marginal utility of Max from windsurfing

| hours per day | total utility from windsurfing | marginal utility from windsurfing |
|---------------|--------------------------------|-----------------------------------|
| 1             | 120                            |                                   |
|               |                                | 100                               |
| 2             | 220                            |                                   |
|               |                                | 80                                |
| 3             | 300                            |                                   |
|               |                                | 60                                |
| 4             | 360                            |                                   |
|               |                                | 36                                |

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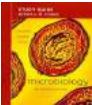
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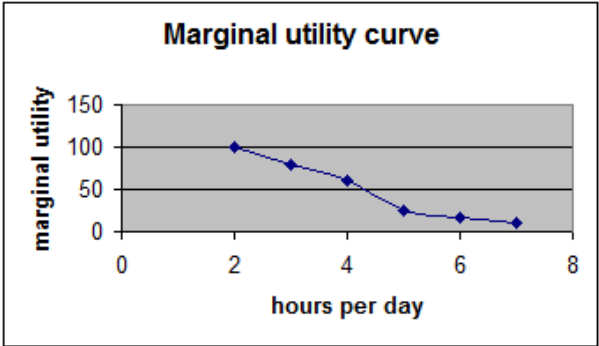
|   |     |    |
|---|-----|----|
|   |     | 16 |
| 6 | 412 |    |
|   |     | 10 |
| 7 | 422 |    |

Comment

Step 4 of 4

Principle of diminishing marginal utility tells that marginal utility decreases as the consumption increases. That is marginal utility form second unit is grater than third unit and so on. This implies that total utility increases as consumption increases but at a decreasing rate.

In the above table derived marginal utility is decreasing as consumption is creasing, i.e.  $MU_2 > MU_3 > MU_4 > MU_5 > MU_6 > MU_7$ . Thus marginal utility from windsurfing obeys the principle of diminishing marginal utility. This is shown below:



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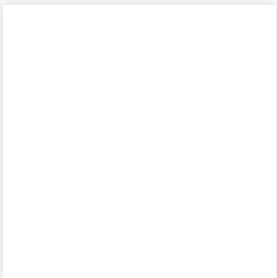
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Chapter 8, Solution 5SPA

The following is the data showing the total utility of windsurfing, snorkeling and marginal utility of snorkeling and...

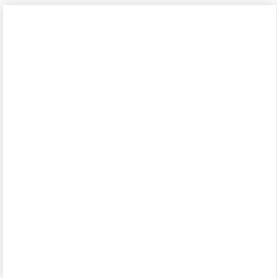
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Chapter 8, Solution 17APA

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## Step-by-step solution

### Step 1 of 3

Marginal utility from snorkeling can be calculated by the same method as we use for windsurfing. That is

$$MU_x = \frac{\Delta U(X)}{\Delta X}$$

Comment

### Step 2 of 3

This is given in the following table:

| Hours per day | total utility from snorkeling | Marginal Utility from snorkeling |
|---------------|-------------------------------|----------------------------------|
| 1             | 40                            |                                  |
|               |                               | 36                               |
| 2             | 76                            |                                  |
|               |                               | 30                               |
| 3             | 106                           |                                  |
|               |                               | 22                               |
| 4             | 128                           |                                  |
|               |                               | 12                               |
| 5             | 140                           |                                  |
|               |                               | 10                               |
| 6             | 150                           |                                  |
|               |                               | 8                                |
| 7             | 158                           |                                  |

Comment

### Step 3 of 3

In the above table this is also shown that marginal utility is decreasing. That is  $MU_2 > MU_3 > MU_4 > MU_5 > MU_6 > MU_7$ . Thus marginal utility from snorkeling also obeys principle of marginal utility. This is shown in the following diagram:

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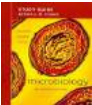
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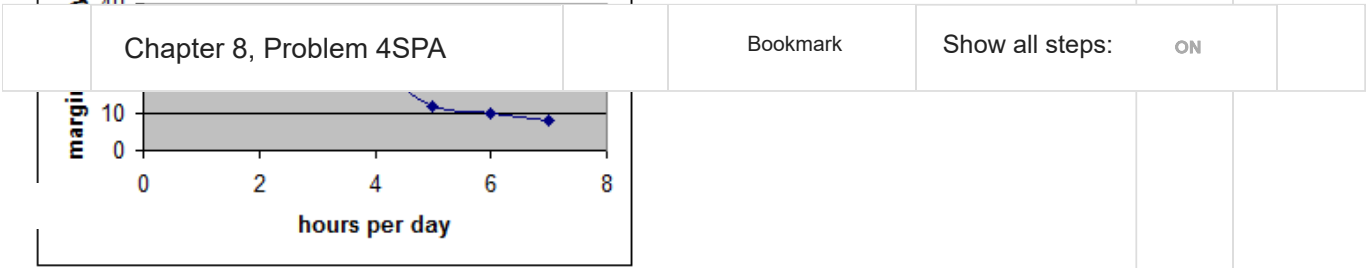


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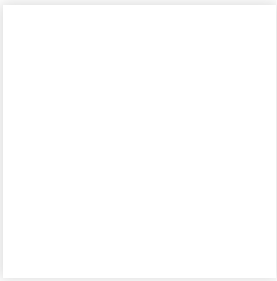
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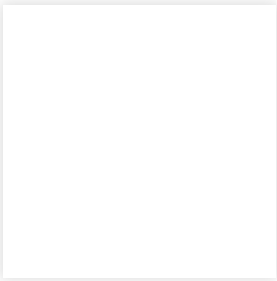
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Cindy has to spend in a month. She spends her income on playing golf and tennis. The price of an hour of golf is and the...

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# Economics | (12th Edition)

## Step-by-step solution

### Step 1 of 2

The following is the data showing the total utility of windsurfing, snorkeling and marginal utility of snorkeling and windsurfing:

| Hours per day | Total Utility from windsurfing | Marginal utility from windsurfing | Total utility from snorkeling | Marginal utility from snorkeling |
|---------------|--------------------------------|-----------------------------------|-------------------------------|----------------------------------|
| 1             | 120                            |                                   | 40                            |                                  |
|               |                                | 100                               |                               | 36                               |
| 2             | 220                            |                                   | 76                            |                                  |
|               |                                | 80                                |                               | 30                               |
| 3             | 300                            |                                   | 106                           |                                  |
|               |                                | 60                                |                               | 22                               |
| 4             | 360                            |                                   | 128                           |                                  |
|               |                                | 36                                |                               | 12                               |
| 5             | 396                            |                                   | 140                           |                                  |
|               |                                | 16                                |                               | 10                               |
| 6             | 412                            |                                   | 150                           |                                  |
|               |                                | 10                                |                               | 8                                |
| 7             | 422                            |                                   | 158                           |                                  |

Total utility from 6<sup>th</sup> hour of windsurfing is 412 and marginal utility is 16. On the other hand total utility from 6<sup>th</sup> hour of snorkeling is 150 and marginal utility is 10. But to consume 16 unit more utility from 6<sup>th</sup> hour of windsurfing Max have to pay \$10, per hour price of renting equipment for windsurfing. Then marginal utility per dollar from 6<sup>th</sup> hour of windsurfing would be marginal utility by price that is  $\frac{16}{10} = 1.6$ .

Comment

### Step 2 of 2

On the other hand marginal utility per dollar from 6<sup>th</sup> hour of snorkeling is  $\frac{10}{5} = 2$ . This implies that per dollar consumes more utility from snorkeling than windsurfing. This result comes as the price of windsurfing equipment is higher than snorkeling. High marginal utility and high price results low marginal utility per dollar. Hence Max enjoys more from 6<sup>th</sup> hour of snorkeling.

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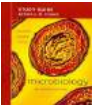
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## Step-by-step solution

### Step 1 of 5

Max's has \$35 and per hour renting rate of windsurfing equipment and snorkeling equipment are \$10 and \$5 respectively. He can rent various combinations of hours and those must satisfy his budget. Suppose there are two goods, say  $X_1$  and  $X_2$ . Budget line is represented by  $P_1X_1+P_2X_2=m$ , where  $P_1$  and  $P_2$  are price of  $X_1$  and  $X_2$ , and  $m$  is income.

Comment

### Step 2 of 5

For example substituting the options (0,7) in the equation;

$$P_1X_1+P_2X_2=m$$

Comment

### Step 3 of 5

Then;  
P<sub>1</sub>= \$10; P<sub>2</sub>= \$5  
X<sub>1</sub>= 0; X<sub>2</sub>=7 and m= 35

Comment

### Step 4 of 5

$$m = P_1X_1 + P_2X_2$$
$$35 = \$10(0) + \$5(7)$$
$$35 = 35$$

Comment

### Step 5 of 5

In the same manner, the remaining values are obtained. Following table gives these various combinations of hours spent on windsurfing and snorkeling.

| Hours spent on windsurfing | Hours spent on snorkeling |
|----------------------------|---------------------------|
| 0                          | 7                         |
| 1                          | 5                         |
| 2                          | 3                         |
| 3                          | 1                         |

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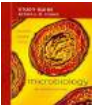
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# Economics | (12th Edition)

## Step-by-step solution

### Step 1 of 3

Following table gives the various combinations of hours spent on windsurfing and snorkeling with the available budget of \$35.

| Hours spent on windsurfing | Hours spent on snorkeling |
|----------------------------|---------------------------|
| 0                          | 7                         |
| 1                          | 5                         |
| 2                          | 3                         |
| 3                          | 1                         |

Adding two columns in the above table, marginal utility per dollar from windsurfing and snorkeling can be listed. For this at first derivation of marginal utility of associated units for both goods is needed. For windsurfing Max's affordable hours are 0, 1, 2, and 3. Total utility for 0 hour is 0 units. For 1 hour marginal utility is 120. For 2 hours Max's marginal utility is total utility from 2 hours minus total utility from 1 hour, i.e. 100 units. Thus marginal utility is calculated by subtracting total utility of previous unit from current unit. Similarly for 3 hours marginal utility is 80 units.

Comment

### Step 2 of 3

In case of snorkeling Max's affordable hours are 7, 5, 3, and 1. For 7 hours Max's marginal utility is total utility from 7 hours minus total utility from 6 hours, i.e. 8 units. Similarly marginal utility from 5 and 3 hours are 12 and 30 units respectively. And For 1 hour marginal utility is 40 units.

Comment

### Step 3 of 3

Marginal utility can be calculated by dividing the marginal utility of a certain unit by its price. Hence we divide all the above marginal utility by its price. The prices of windsurfing and snorkeling are \$10 and \$5 respectively. Now marginal utility per dollar is given bellow:

| Hours spent on windsurfing | Marginal utility per dollar from windsurfing | Marginal utility per dollar from snorkeling | Hours spent on snorkeling |
|----------------------------|----------------------------------------------|---------------------------------------------|---------------------------|
| 0                          | 0                                            | 1.6                                         | 7                         |
| 1                          | 12                                           | 2.4                                         | 5                         |
| 2                          | 10                                           | 6                                           | 3                         |
| 3                          | 8                                            | 8                                           | 1                         |

Thus the above table give the MU/\$ for windsurfing and snorkeling.

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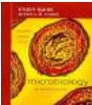
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Chapter 8, Problem 8SPA

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Step-by-step solution

Step 1 of 3

a)

Make a table that shows the various combinations of hours spent windsurfing and snorkeling that Max can afford with his \$35 budget.

This can be seen as below:

| Hours windsurfing | MU per dollar from windsurfing | Hours snorkeling | MU per dollar from snorkeling |
|-------------------|--------------------------------|------------------|-------------------------------|
| 3                 | $(300-220)/10 = 8$             | 1                | $(40-5)/5 = 8$                |
| 2                 | $(220-120)/10 = 10$            | 3                | $(106-76)/5 = 6$              |
| 1                 | $(120-0)/10 = 12$              | 5                | $(140-128)/5 = 2.4$           |
| 0                 |                                | 7                | $(158-150)/5 = 1.6$           |

Utility is maximized at the point MU per dollar from windsurfing equals MU per dollar from snorkeling.

This occurs when M windsurfs for 3 hour and snorkels for 1 hour.

Comment

Step 2 of 3

b)

If M windsurfs another extra hour, he pays \$10 and gains 60 units of utility (the marginal utility from the 4th hour), which is 6.0 units of utility per dollar. So if he spends a dollar more on windsurfing, his utility from windsurfing increases by 6.0.

At the same time, if he spends an hour less on snorkeling, he saves \$5 and loses 40 units of utility (the marginal utility from the 1st hour of snorkeling), which is 8.0 units of utility per dollar. So if he spends a dollar less on snorkeling, he loses 8.0 units of utility.

Thus, overall, spending a dollar more on windsurfing and a dollar less on snorkeling will lower M's total utility by 2.0 units of utility.

Comment

Step 3 of 3

c)

If M snorkels another hour, he pays \$5 and gains 36 units of utility (the marginal utility from the 2nd hour), which is 7.2 units of utility per dollar. So if he spends a dollar more on snorkeling, his utility from snorkeling increases by 7.2.

At the same time, If he spends an hour less on windsurfing, he saves \$10 and loses 80 units of utility (the marginal utility from the 3rd hour of windsurfing), which is 8.0 units of utility per dollar. So if he spends a dollar less on windsurfing, he loses 8.0 units of utility.

Thus, overall, spending a dollar more on snorkeling and a dollar less on windsurfing lowers M's total utility by 0.8 units of utility.

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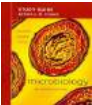
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## Step-by-step solution

### Step 1 of 6

Demand curve indicates the quantity of a commodity that will be bought at a given price during a particular period of time. Demand curve is downward sloping, because price and quantity demanded are negatively related, i.e. if price raises quantity demand falls and vice versa. Thus Max demands more at price \$5 than the high price \$10. He has \$35 to spend in a day on windsurfing and snorkeling. The first per hour renting prices of windsurfing and snorkeling equipment's were \$10 and \$5 respectively. The affordable combinations of time and best bundle are determined below:

| Hours per day on windsurfing | Total utility from windsurfing | Total utility from windsurfing and snorkeling | Total utility from snorkeling | Hours per day on snorkeling |
|------------------------------|--------------------------------|-----------------------------------------------|-------------------------------|-----------------------------|
| 0                            | 0                              | 158                                           | 158                           | 7                           |
| 1                            | 120                            | 260                                           | 140                           | 5                           |
| 2                            | 220                            | 326                                           | 106                           | 3                           |
| 3                            | 300                            | 340                                           | 40                            | 1                           |

In the above table among all affordable combinations of hours Max gets maximum utility from 3 hours per day on windsurfing and 1 hour per day on snorkeling.

Thus when renting price of windsurfing equipments is \$10, Max demands 3 hours per day on windsurfing.

Comment

### Step 2 of 6

When price of renting of windsurfing equipment is \$5 affordable combinations also change. Max demands more hours on windsurfing. This is given below:

| Hours per day on windsurfing | Total utility from windsurfing | Total utility from windsurfing and snorkeling | Total utility from snorkeling | hours per day on snorkeling |
|------------------------------|--------------------------------|-----------------------------------------------|-------------------------------|-----------------------------|
| 0                            | 0                              | 158                                           | 158                           | 7                           |
| 1                            | 120                            | 270                                           | 150                           | 6                           |
| 2                            | 220                            | 360                                           | 140                           | 5                           |
| 3                            | 300                            | 428                                           | 128                           | 4                           |
| 4                            | 360                            | 466                                           | 106                           | 3                           |
| 5                            | 396                            | 472                                           | 76                            | 2                           |
| 6                            | 412                            | 452                                           | 40                            | 1                           |
| 7                            | 422                            | 422                                           | 0                             | 0                           |

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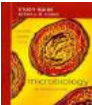
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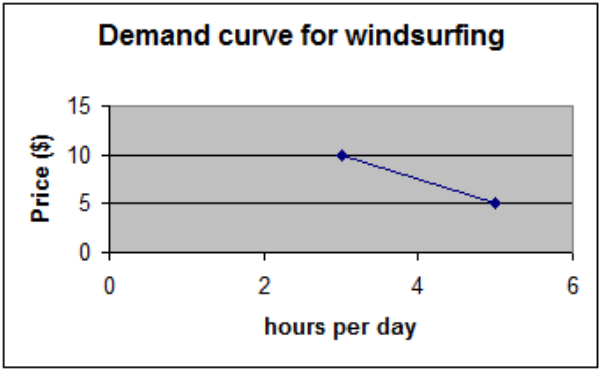
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Step 3 of 6

Demand curve of Max for windsurfing can be drawn from the above two tables, as max demands 3 hours at price \$10 and 5 hours at price \$5 . This is drawn in the following diagram:



Comment

Step 4 of 6

Elasticity of demand is a response of demand for a product to a change in price. It may be expressed mathematically by dividing the percentage change in demand by the percentage change in price. It is given in below:

$$\begin{aligned}\epsilon_D &= \frac{\% \Delta Q}{\% \Delta P} \\ &= \frac{\frac{\Delta Q}{Q} \times 100}{\frac{\Delta P}{P} \times 100} \\ &= \frac{\Delta Q}{Q} \times \frac{P}{\Delta P} \\ &= \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}\end{aligned}$$

Description of variables:

- $\epsilon_D$  = elasticity of demand
- $\Delta Q$  = change in quantity
- $\Delta P$  = change in price
- $Q$  = quantity before price change
- $P$  = price after changed.

Comment

Step 5 of 6

In this problem:

$$\begin{aligned}\Delta Q &= Q_2 - Q_1 \\ &= 5 - 3 \\ &= 2\end{aligned}$$

And

$$\begin{aligned}\Delta P &= P_2 - p_1 \\ &= 5 - 10 \\ &= -5\end{aligned}$$

But elasticity always takes absolute value. So,  $\Delta P = 5$

Comment

Step 6 of 6

Then

$$\begin{aligned}\epsilon_D &= \frac{2}{5} \times \frac{5}{5} \\ &= \frac{2}{5} \\ &= 0.4\end{aligned}$$

Therefore  $0 < 0.4 < 1$ , it implies that elasticity of demand is relatively inelastic. That means percentage change in quantity is less than the percentage change in price.

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Step-by-step solution

Step 1 of 6

Price change of a goods affects the demand of any substitute or complement of its. According to law of demand price and quantity are negatively related, assuming all other factors constant. But quantity demand of a good depends on its own price as well as other good's price and income.

When the renting price of windsurfing equipments was **\$10** per hour, Max demands 3 hours per day on windsurfing and 1 hour per day on snorkeling. This combination of 3 hours and 1 hour is derived by total utility maximization. Given the renting price of windsurfing and snorkeling equipments and income Max has some affordable combinations of hours that he can spends on windsurfing and snorkeling. Then he chooses combinations of hours that he can spend on windsurfing and snorkeling separately and total utility achieves its maximum.

Comment

Step 2 of 6

Following table is given in respects of above judgments:

| Graph the demand supply | Graph the demand supply | Graph the demand supply | Graph the demand supply | Graph the demand supply |
|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| 0                       | 0                       | 158                     | 158                     | 7                       |
| 1                       | 120                     | 260                     | 140                     | 5                       |
| 2                       | 220                     | 326                     | 106                     | 3                       |
| <b>3</b>                | <b>300</b>              | <b>340</b>              | <b>40</b>               | <b>1</b>                |

Extreme two columns indicate affordable combinations of hours per day. Among these 3 hours per day on windsurfing and 1 hour per day on snorkeling makes Max's total utility highest.

Comment

Step 3 of 6

Now consider the case when renting price of windsurfing equipments falls to **\$5** . With the extra **\$5** per hour he can spend on windsurfing or snorkeling so that his utility achieves its maximum. Following tables gives information about efficient combinations of hours:

| Hours per day on windsurfing | Total utility from windsurfing | Total utility from windsurfing and snorkeling | Total utility from snorkeling | hours per day on snorkeling |
|------------------------------|--------------------------------|-----------------------------------------------|-------------------------------|-----------------------------|
| 0                            | 0                              | 158                                           | 158                           | 7                           |
| 1                            | 120                            | 270                                           | 150                           | 6                           |
| 2                            | 220                            | 360                                           | 140                           | 5                           |
| 3                            | 300                            | 428                                           | 128                           | 4                           |
| 4                            | 360                            | 466                                           | 106                           | 3                           |
| <b>5</b>                     | <b>396</b>                     | <b>472</b>                                    | <b>76</b>                     | <b>2</b>                    |

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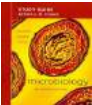
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In the above table it is shown that spending on 5 hours per day on windsurfing and 2 hours per day on snorkeling Max attains its maximum utility.

Therefore fall in the price of windsurfing equipments affects the demands of both goods positively.

Comment

Step 4 of 6

**Cross elasticity of demand** measures the responsiveness in the quantity demand of one good when a change in price takes place in another good. More specifically cross elasticity of demand is percentage change in quantity of a good due to percentage change in price of another good.

Now, Max's cross elasticity of demand of snorkeling with respect to the price windsurfing refers to the change in quantity demand of snorkeling due to change in the price of windsurfing.

Comment

Step 5 of 6

Mathematically,

Cross elasticity of demand =  $\frac{\Delta Q_s}{\Delta P_w} \times \frac{P_{ow}}{Q_{os}}$

$\Delta Q_s$  = percentage change in quantity demand of snorkeling

$\Delta P_w$  = percentage change in price of windsurfing

$P_{ow}$  = old price of windsurfing

$Q_{os}$  = old quantity demand of snorkeling.

When price of windsurfing equipment was \$10 demand for snorkeling equipments was 1hour per day, and when price changes to \$5 then demand for snorkeling equipments changes to 2 hours per day.

Now we derive

$\Delta Q_s = 2 - 1$   
 $= 1$

And

$\Delta P_w = 5 - 10$   
 $= -5$

Now cross elasticity =  $\frac{1}{-5} \times \frac{10}{1}$

= -2

Therefore cross elasticity for Max is less than zero.

Comment

Step 6 of 6

To answer the question about substitute or complement, we use the following rule of thumb:

If cross elasticity of demand  $> 0$  then the goods are substitutes.

If cross elasticity of demand  $= 0$  , then the goods are independent.

If cross elasticity of demand  $< 0$  , then the goods are complements.

Here cross elasticity of demand is -2, which is less than zero. Therefore windsurfing and snorkeling are complements. That means with the decrease in price of windsurfing equipments the demand of windsurfing equipments as well as snorkeling equipments increase.

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## Step-by-step solution

### Step 1 of 6

Generally with an increase income demands for the goods also increase. These goods are normal goods. But when demand of the goods decreases due to increase in income; theses goods are called inferior goods.

Initial Max has \$35 to spend on windsurfing and snorkeling equipments per day. Given that renting price of windsurfing and snorkeling equipments are \$10 and \$5 respectively.

Comment

### Step 2 of 6

Given these prices and income Max can choose any combination of these two goods. The combinations are given below:

| Hours per day<br>on windsurfing | Hours per day<br>on snorkeling |
|---------------------------------|--------------------------------|
| 0                               | 7                              |
| 1                               | 5                              |
| 2                               | 3                              |
| 3                               | 1                              |

Comment

### Step 3 of 6

But he only chooses that combination which gives him highest utility. Derivation is given in the following diagram:

| Hours per day<br>on windsurfing | Total utility<br>from<br>windsurfing | Total utility from<br>windsurfing and<br>snorkeling | Total utility<br>from<br>snorkeling | Hours per day<br>on snorkeling |
|---------------------------------|--------------------------------------|-----------------------------------------------------|-------------------------------------|--------------------------------|
| 0                               | 0                                    | 158                                                 | 158                                 | 7                              |
| 1                               | 120                                  | 260                                                 | 140                                 | 5                              |
| 2                               | 220                                  | 326                                                 | 106                                 | 3                              |
| 3                               | 300                                  | 340                                                 | 40                                  | 1                              |

From the above table it can be shown that from 3 hours on windsurfing and 1 hour on snorkeling Max can get highest utility. Hence Max’s demand for windsurfing is 3 hours per day.

Comment

### Step 4 of 6

Now, consider income increases to \$55 but the prices remain same. His affordable combinations of windsurfing and snorkeling will also change. This is taken into account that Max can spend maximum 7 hours both on windsurfing and snorkeling. This is given below:

|  |  |
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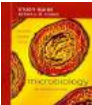
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|   |                          | 3 |          |                    |  |
| 5 |                          | 1 |          |                    |  |

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Step 5 of 6

Following table is given below to derive highest utility combination:

| Hours per day on windsurfing | Total utility from windsurfing | Total utility from windsurfing and snorkeling | Total utility from snorkeling | Hours per day on snorkeling |
|------------------------------|--------------------------------|-----------------------------------------------|-------------------------------|-----------------------------|
| 2                            | 220                            | 378                                           | 158                           | 7                           |
| 3                            | 300                            | 440                                           | 140                           | 5                           |
| 4                            | 360                            | 466                                           | 106                           | 3                           |
| 5                            | 396                            | 436                                           | 40                            | 1                           |

From the above table it can be said that highest utility is derived by 4 hours on windsurfing and 3 hours on snorkeling per day. It is seen that demands for both goods are increased due to increase in income. With higher income Max demands 4 hours per day on windsurfing.

Comment

Step 6 of 6

The demand for windsurfing in respect to income is given below:

| Hours on windsurfing | Income |
|----------------------|--------|
| 3                    | \$35   |
| 4                    | \$55   |

Since quantity demand for windsurfing is positively related with income, as income rises demand for windsurfing increases, Windsurfing is normal good.

Comment

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Recommended solutions for you in Chapter 8

Chapter 8, Solution 5SPA

The following is the data showing the total utility of windsurfing, snorkeling and marginal utility of snorkeling and...

See solution



Chapter 8, Solution 17APA

Cindy has to spend in a month. She spends her income on playing golf and tennis. The price of an hour of golf is and the...

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Chapter 8, Problem 13SPA

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Step-by-step solution

Step 1 of 6

Generally with an increase income demands for the goods also increase. These goods are normal goods. But when demand of the goods decreases due to increase in income; theses goods are called inferior goods.

Initial Max has \$35 to spend on windsurfing and snorkeling equipments per day. Given that renting price of windsurfing and snorkeling equipments are \$10 and \$5 respectively.

Comment

Step 2 of 6

Given these prices and income Max can choose any combination of these two goods. The combinations are given below:

| Hours per day on windsurfing | Hours per day on snorkeling |
|------------------------------|-----------------------------|
| 0                            | 7                           |
| 1                            | 5                           |
| 2                            | 3                           |
| 3                            | 1                           |

Comment

Step 3 of 6

But he only chooses that combination which gives him highest utility. Derivation is given in the following diagram:

| Hours per day on windsurfing | Total utility from windsurfing | Total utility from windsurfing and snorkeling | Total utility from snorkeling | Hours per day on snorkeling |
|------------------------------|--------------------------------|-----------------------------------------------|-------------------------------|-----------------------------|
| 0                            | 0                              | 158                                           | 158                           | 7                           |
| 1                            | 120                            | 260                                           | 140                           | 5                           |
| 2                            | 220                            | 326                                           | 106                           | 3                           |
| 3                            | 300                            | 340                                           | 40                            | 1                           |

From the above table it can be shown that from 3 hours on windsurfing and 1 hour on snorkeling Max can get highest utility. Hence Max’s demand for snorkeling is 1 hour per day.

Comment

Step 4 of 6

Now, consider income increases to \$55 but the prices remain same. His affordable combinations of windsurfing and snorkeling will also change. This is taken into account that Max can spend maximum 7 hours both on windsurfing and snorkeling. This is given below:

| Hours per day on windsurfing | Hours per day on snorkeling |
|------------------------------|-----------------------------|
|------------------------------|-----------------------------|

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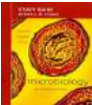
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| 3 | 5 |
| 4 | 3 |
| 5 | 1 |

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Step 5 of 6

Following table is given below to derive highest utility combination:

| Hours per day on windsurfing | Total utility from windsurfing | Total utility from windsurfing and snorkeling | Total utility from snorkeling | Hours per day on snorkeling |
|------------------------------|--------------------------------|-----------------------------------------------|-------------------------------|-----------------------------|
| 2                            | 220                            | 378                                           | 158                           | 7                           |
| 3                            | 300                            | 440                                           | 140                           | 5                           |
| 4                            | 360                            | 466                                           | 106                           | 3                           |
| 5                            | 396                            | 436                                           | 40                            | 1                           |

From the above table it can be said that highest utility is derived by 4 hours on windsurfing and 3 hours on snorkeling per day. It is seen that demands for both goods are increased due to increase in income. With higher income Max demands 3 hours per day on snorkeling.

[Comment](#)

Step 6 of 6

The demand for snorkeling in respect to income is given below:

| Hours on snorkeling | Income |
|---------------------|--------|
| 1                   | \$35   |
| 3                   | \$55   |

Since quantity demand for snorkeling is positively related with income, as income raises demand for snorkeling also increases, hence snorkeling is a normal good.

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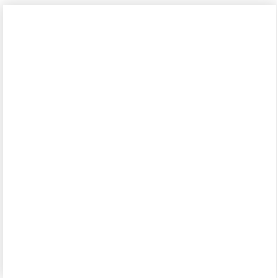
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Chapter 8, Solution 5SPA

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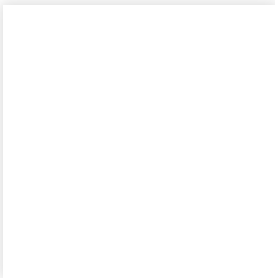
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Chapter 8, Solution 17APA

Cindy has to spend in a month. She spends her income on playing golf and tennis. The price of an hour of golf is and the...

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Step-by-step solution

Step 1 of 3

Cindy has **\$70** to spend in a month. She spends her income on playing golf and tennis. The price of an hour of golf is **\$10** and the price of an hour of tennis is **\$5**. She has various affordable combinations of hours playing golf and tennis. All these combinations spend her entire income, i.e. **\$70**, and she remains on her budget.

Comment

Step 2 of 3

Following table shows Cindy's affordable combinations:

| Hours on playing golf | Expenditure for golf (\$) | Hours on playing tennis | Expenditure for tennis (\$) | Total expenditure for golf and tennis (\$) |
|-----------------------|---------------------------|-------------------------|-----------------------------|--------------------------------------------|
| 1                     | 10                        | 12                      | 60                          | 70                                         |
| 2                     | 20                        | 10                      | 50                          | 70                                         |
| 3                     | 30                        | 8                       | 40                          | 70                                         |
| 4                     | 40                        | 6                       | 30                          | 70                                         |
| 5                     | 50                        | 4                       | 20                          | 70                                         |
| 6                     | 60                        | 2                       | 10                          | 70                                         |

All these combinations of hours exhaust her entire income.

Comment

Step 3 of 3

Now Cindy has decided to spend **\$100** in a month playing on golf and tennis. So she increases her expenditure from **\$70** to **\$100**. Thus her consumption possibilities also change so that she can exhaust all her income. These are given in the following table:

| Hours on playing golf | Expenditure for golf (\$) | Hours on playing tennis | Expenditure for tennis (\$) | Total expenditure for golf and tennis (\$) |
|-----------------------|---------------------------|-------------------------|-----------------------------|--------------------------------------------|
| 1                     | 10                        | 18                      | 90                          | 100                                        |
| 2                     | 20                        | 16                      | 80                          | 100                                        |
| 3                     | 30                        | 14                      | 70                          | 100                                        |
| 4                     | 40                        | 12                      | 60                          | 100                                        |
| 5                     | 50                        | 10                      | 50                          | 100                                        |
| 6                     | 60                        | 8                       | 40                          | 100                                        |
| 7                     | 70                        | 6                       | 30                          | 100                                        |

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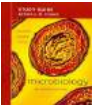
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|---|----|---|----|-----|
| 9 | 90 | 2 | 10 | 100 |
|---|----|---|----|-----|

Hence when income increases Cindy has more consumption possibilities compared to the first case when income was **\$70**.

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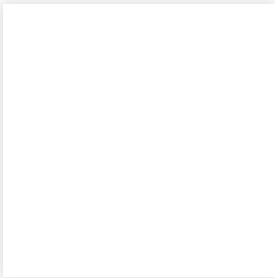
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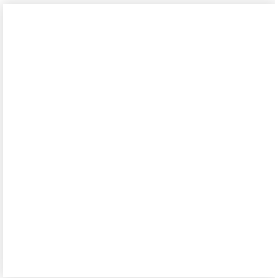
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## Step-by-step solution

### Step 1 of 3

The given information proffers that the income earned by Cindy is \$70. With a price of \$10 per hour of Tennis, the maximum number of hours she can play Tennis is 7 (and no golf). Similarly, she can play for 7 hours Golf (and no Tennis) with her given income.

Comment

### Step 2 of 3

A bundle is affordable when it can satisfy the budget constraint:

$$M \geq xPx + yPy$$

where

$M = \text{income}$

$x = \text{good } x$

$y = \text{good } y$

$Px = \text{price of } x$

$Py = \text{price of } y$

Substitute the values of the consumption in the desirable bundle, price of two goods and the given income to check whether the desirable bundle is affordable.

$$70 \geq xPx + yPy$$

$$70 \geq T \times 10 + G \times 10$$

$$7 \geq T + G$$

Hence the budget constraint suggests that all bundles of hours spent on Tennis and Golf that satisfy  $7 \geq T + G$  , will be affordable. The affordable bundles are: 0 hour of Tennis and 7 hours of Golf, 1 hour of Tennis and 6 hours of Golf, 2 hours of Tennis and 5 hours of Golf, 3 hours of Tennis and 4 hours of Golf, 4 hours of Tennis and 3 hours of Golf, 5 hours of Tennis and 2 hours of Golf, 6 hours of Tennis and 1 hour of Golf and 7 hours of Tennis and 0 hour of Golf.

Comment

### Step 3 of 3

Data related to the change in the utilities and its effect on optimal consumption bundle chosen by Cindy is encapsulated in Table 1.

| Hours at Golf at \$10 each    | Hours at Tennis at \$10 each                                |                                                                |                                 |                                                               |                                                                |
|-------------------------------|-------------------------------------------------------------|----------------------------------------------------------------|---------------------------------|---------------------------------------------------------------|----------------------------------------------------------------|
| Number of Hours spent at Golf | Marginal Utility from Last Hours at Golf (MU <sub>G</sub> ) | Marginal Utility per Dollar (MU <sub>G</sub> /P <sub>G</sub> ) | Number of Hours spent at Tennis | Marginal Utility from Last Hours at Tennis (MU <sub>T</sub> ) | Marginal Utility per Dollar (MU <sub>T</sub> /P <sub>T</sub> ) |
| 1                             | 80                                                          | 8                                                              | 6                               | 2                                                             | 0.2                                                            |
| 2                             | 60                                                          | 6                                                              | 5                               | 5                                                             | 0.5                                                            |
| 3                             | 40                                                          | 4                                                              | 4                               | 10                                                            | 1                                                              |
| 4                             | 30                                                          | 3                                                              | 3                               | 30                                                            | 3                                                              |
| 5                             | 20                                                          | 2                                                              | 2                               | 36                                                            | 3.6                                                            |
| 6                             | 10                                                          | 1                                                              | 1                               | 40                                                            | 4                                                              |
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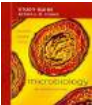
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Table 1

Investigate the table to note that the marginal utility per dollar spent on last hour of Tennis played is equal to the marginal utility per dollar spent on last hour of Golf played when Cindy chooses 3 hours of Tennis and 4 hours of Golf. This combination is on the budget line ( $\$10\times3 + \$10\times4 = \$70$ ).

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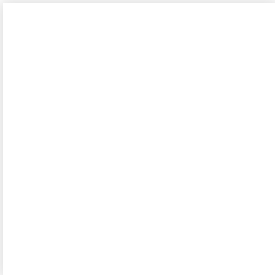
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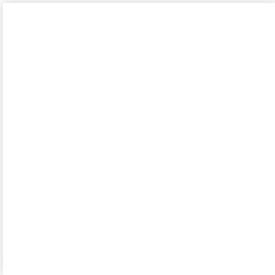
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### Chapter 8, Solution 17APA

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Step-by-step solution

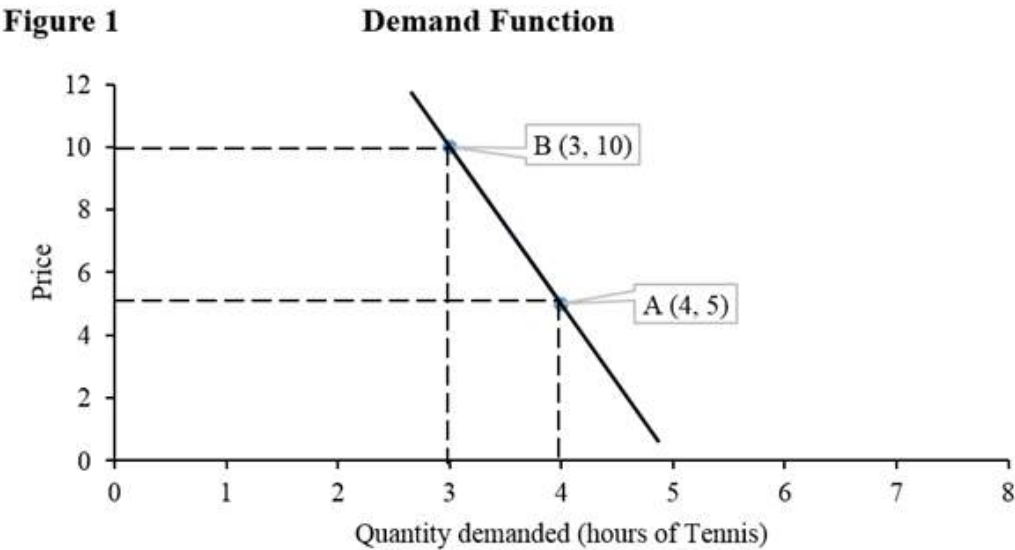
Step 1 of 2

The demand for a particular product represents a downward sloping curve showing a negative relationship between quantity demanded and price. This implies a fall in the quantity demanded when the price increases and vice versa. Note that in this scenario, there is an expansion of contraction along the demand curve.

Comment

Step 2 of 2

The demand curve for Cindy is drawn in Figure 1. From her utility maximization choice when the price of playing one hour of Tennis was \$5, observe that the optimal choice was 4 hours of Tennis.



As soon as the price of playing one hour of Tennis increases from \$5 to \$10, the optimal choice falls from 4 hours of Tennis to 3 hours. Note that the percentage increase in the price is 100 percent while quantity demanded falls by 25 percent. Hence, the demand is inelastic.

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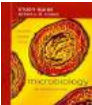
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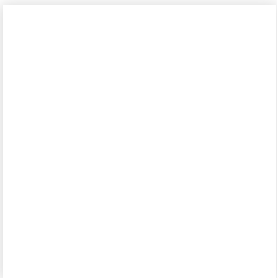
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Chapter 8, Solution 5SPA

The following is the data showing the total utility of windsurfing, snorkeling and marginal utility of snorkeling and...

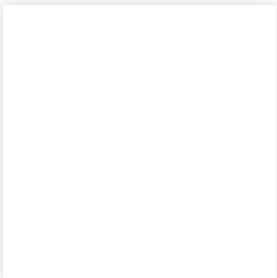
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Chapter 8, Solution 17APA

Cindy has to spend in a month. She spends her income on playing golf and tennis. The price of an hour of golf is and the...

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Chapter 8, Problem 23APA

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Step-by-step solution

Step 1 of 2

The given information entails that the income earned by Cindy is \$70. With a price of \$5 per hour of Tennis, the maximum number of hours she can play Tennis is 14 (and no golf). Similarly, she can play for 7 hours Golf (and no Tennis) with her given income. When her income falls to \$35, she can now play a maximum of 7 hours of Tennis or 3 and a half hours of Golf.

Comment

Step 2 of 2

With this change, note that the marginal utility per dollar spent on last hour of Tennis played is equal to the marginal utility per dollar spent on last hour of Golf played when **Cindy chooses 3 hours of Tennis and 2 hours of Golf**. This is valued at 6 utils per dollar (60/10 and 30/5). This combination lies on the budget line ( $\$10 \times 2 + \$5 \times 3 = \$35$ ).

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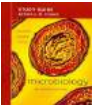
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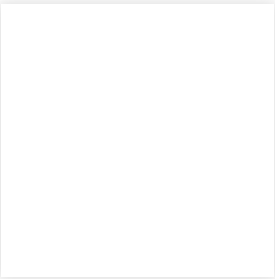
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### Chapter 8, Solution 5SPA

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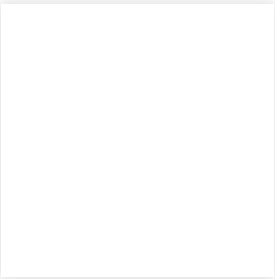
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### Chapter 8, Solution 17APA

Cindy has to spend in a month. She spends her income on playing golf and tennis. The price of an hour of golf is and the...

See solution



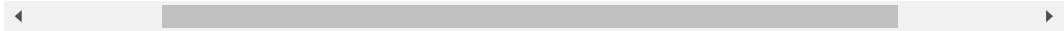
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Step-by-step solution

Step 1 of 4

- a. If Cindy allocates a total of four hours a day to golf and tennis Cindy spends different combinations to these activities. This is given in the following table.

| Combinations | Hours per day on golf | Hours per day on tennis |
|--------------|-----------------------|-------------------------|
| A            | 1                     | 3                       |
| B            | 2                     | 2                       |
| C            | 3                     | 1                       |



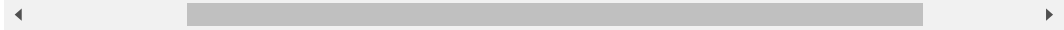
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Step 2 of 4

Cindy plays that much of time on golf and tennis, which maximizes her total utility. Particular in this case where price does not matter, Utility maximization condition states equality between marginal utility of golf and tennis. This is derived in the following table.

| Combinations | Hours per day on golf | Marginal utility from golf | Hours per day on tennis | Marginal utility from tennis |
|--------------|-----------------------|----------------------------|-------------------------|------------------------------|
| A            | 1                     | 80                         | 3                       | 30                           |
| B            | 2                     | 60                         | 2                       | 36                           |
| C            | 3                     | 40                         | 1                       | 40                           |

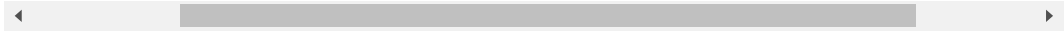
Marginal utility of golf and tennis equal in the last combination. Hence Cindy plays 3 hours per day on golf and 1 hour per day on tennis.



Comment

Step 3 of 4

- b. In the above table marginal utility from golf and tennis has been given. Cindy chooses that combination of hours which give her maximum utility. That is combination C. Hence Cindy's marginal utility from golf is 40 units and from tennis is 40 units also.



Comment

Step 4 of 4

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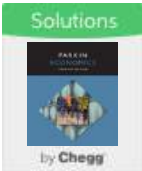
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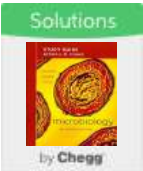
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from golf and tennis are  $MU_G$  and  $MU_T$  respectively and the prices are  $P_G$  and  $P_T$  respectively, then for equilibrium:

<

$$\frac{MU_G}{P_G} = \frac{MU_T}{P_T}$$

>

If price does not matter, then we can neutralize both prices by canceling just like prices are equal. Then we get:

$$MU_G = MU_T$$

Thus equality between marginal utility from golf and tennis is sufficient. This is possible as price does not affect her preference.

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### Recommended solutions for you in Chapter 8

#### Chapter 8, Solution 5SPA

The following is the data showing the total utility of windsurfing, snorkeling and marginal utility of snorkeling and...

[See solution](#)

$$\frac{16}{10} = 1.6$$

#### Chapter 8, Solution 17APA

Cindy has to spend in a month. She spends her income on playing golf and tennis. The price of an hour of golf is and the...

[See solution](#)

\$70

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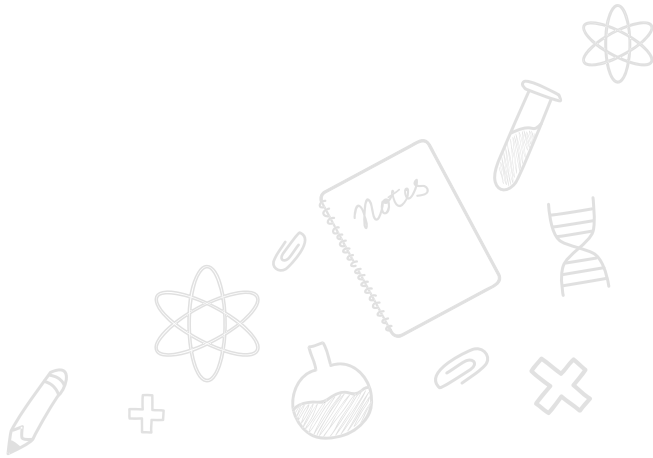
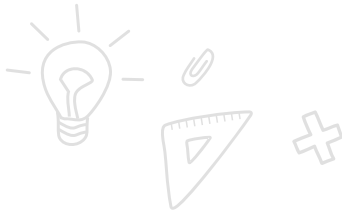
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Step-by-step solution

Step 1 of 6

(a) Ben buys 10,000 gallons of tape water and 2 bunches of flowers in a year. So Ben consumes more water, because it is necessary for daily life and he cannot live without water. On the other hand, flower is not necessary to him, it is a luxury product. Hence water is more valuable to Ben. That's why he consumes more water than flower.

Comment

Step 2 of 6

(b) Though water is high valued product, its price is low compared to flowers. But how this can happened that valuable water is so cheap while relatively useless flowers is so expensive. This is so called *paradox of value*.

Comment

Step 3 of 6

Paradox of value can be resolved by distinguishing between total utility and marginal utility. First of all, a consumer buys a combination of goods that maximize total utility. Total utility is maximized when marginal utility per dollar from goods is equal. Now marginal utility per dollar can be derived by diving marginal utility by price. Ben consumes more water, his utility from water is greater than flowers but the marginal utility is lower. This can be inherited from low of diminishing marginal utility.

Comment

Step 4 of 6

Given that, marginal utility from water is 0.5 units per gallon and he consumes 10,000 gallons of water spending \$50. Then price per unit is. Now marginal utility per dollar can be derived as

$$\begin{aligned} \text{marginal utility per dollar} &= \frac{\text{marginal utility}}{\text{price}} \\ &= \frac{0.5}{0.005} \\ &= 100 \text{units} \end{aligned}$$

Thus marginal utility per dollar from water is 100 units. As Ben consumes 2 bunches of flower along with 10,000 gallons of water to maximize total utility, then marginal utility per dollar from flowers must be also 100 units.

Comment

Step 5 of 6

It is given that Ben consumes 2 bunches of flowers by spending \$50. Then its price would be . From the above analysis we get that marginal utility per dollar from glowers would be 100 units. Now marginal utility can derived by

$$\begin{aligned} \text{marginal utility per dollar} &= \frac{\text{marginal utility}}{\text{price}} \\ \text{marginal utility} &= \text{marginal utility per dollar} \times \text{price} \\ &= 100 \times 25 \\ &= \boxed{2,500} \end{aligned}$$

Thus marginal utility from flowers is 2500.

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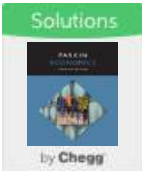
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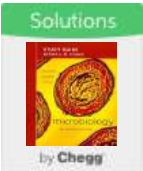
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marginal utility and higher price give lower marginal utility per dollar (100 units) from flowers. That is also equal to marginal utility per dollar from water while lower marginal utility and lower price gives the same marginal utility per dollar as flowers.

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## Recommended solutions for you in Chapter 8

### Chapter 8, Solution 5SPA

The following is the data showing the total utility of windsurfing, snorkeling and marginal utility of snorkeling and...

See solution

$$\frac{16}{10} = 1.6$$

### Chapter 8, Solution 17APA

Cindy has to spend in a month. She spends her income on playing golf and tennis. The price of an hour of golf is and the...

See solution

\$70

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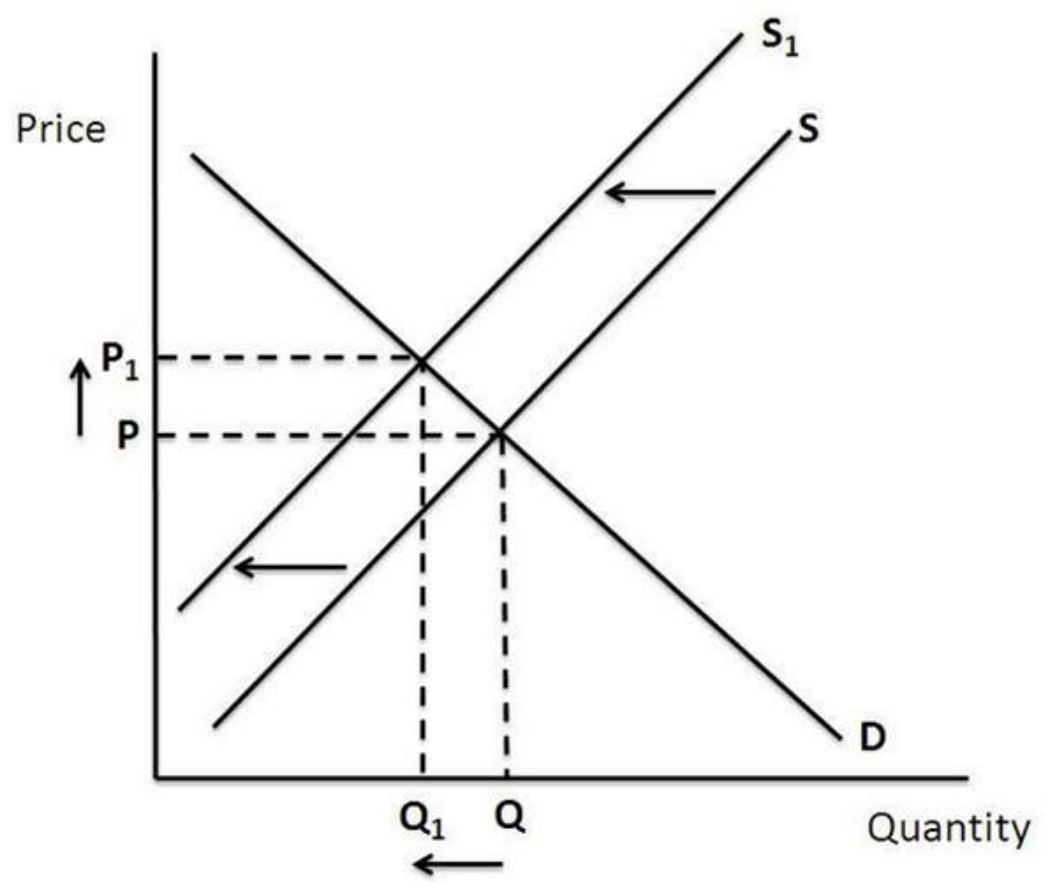
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Step-by-step solution

Step 1 of 4

- (a)
- (i)
- When sugary drinks are banned at restaurants, there supply falls drastically, thereby shifting the supply curve to the left.
- Keeping demand curve fixed, this leads to an increase in price and decrease in quantity of sugary drinks in the market.
- This can be seen as below:



Comment

Step 2 of 4

- (ii)
- Following the law of demand, an increase in price of these drinks makes consumers reduce their demand as it becomes costly for them to consume such drinks.

Comment

Step 3 of 4

- (b)
- (i)
- As tax is imposed on sugary drinks, the marginal utility of a sugary drink decreases because consumers will now have to pay more to get additional drinks and thus their additional utility will start declining.

Comment

Step 4 of 4

- (ii)
- As prices increase, consumer surplus will decrease, as it is measured as the difference between consumer willingness to pay for the good and the actual price paid.

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